

## Exercises for Chapter 2.6, with solutions

### Graph Algorithms

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The exercises cover some of the material covered in Chapter 2.6. Each exercise is of one of the following types: basic, suggested, or more challenging. Basic exercises deal with concepts from lectures and are intended to check understanding of basic concepts. Suggested exercises deal with concepts from lectures and are similar to exercises that you may expect on exams. More challenging exercises address concepts from lectures, but these exercises are less likely to appear on exam or are intended for a higher grade.

1. (*Maximum-weight matchings; basic.*)

Give an example of a graph with positive edge weights in which no *maximum-weight* matching is also of *maximum size*.

**Solution:** For a small example, we can already take a path with vertex set  $\{a, b, c, d\}$  and a set of edges  $\{ab, bc, cd\}$ , where  $w(ab) = w(cd) = 1$  and  $w(bc) = 3$ . In this case,  $M = \{bc\}$  is the only maximum-weight matching (with total weight 3). The only maximum matching in the graph is  $\{ab, cd\}$ .

2. (*Matchings in general graphs; suggested.*)

A matching in a graph is *maximal* if it is not contained in any other matching. Let  $M$  be a maximal matching in a graph  $G$ , and let  $M^*$  be a maximum matching in  $G$ . Show that  $|M| \geq \frac{1}{2}|M^*|$ .

**Solution:** Let  $V(M)$  be the set of vertices of the graph  $G$  that are endpoints of some edge of the matching  $M$ . Since the matching  $M$  is maximal, no edge in the graph has both endpoints in the set  $V(G) \setminus V(M)$ . Consequently, each of the matching edges  $M^*$  also has at least one endpoint in the set  $V(M)$ . Since no vertex in the set  $V(M)$  is an endpoint of more than one matching edge  $M^*$ , it follows that  $|M^*| \leq |V(M)| = 2|M|$ .

3. (*Applications of maximum matchings; suggested.*)

Let  $G = (V, E)$  be a graph without isolated vertices. An **edge cover** in a graph  $G$  is a set  $F$  of edges such that every vertex  $v \in V$  is an endpoint of some edge in the set  $F$ . Show that the minimum cardinality of an edge cover in  $G$  is equal to the maximum cardinality of a matching  $M$  in  $G$ , increased by  $|V| - 2|M|$ .

Give a polynomial-time algorithm for computing a minimum edge cover.

**Solution:** Let  $M$  be a maximum matching in the graph  $G = (V, E)$  and let  $X$  be the set of vertices that are endpoints of some edge of  $M$ . Then no two vertices  $u$  and  $v$  in the set  $V \setminus X$  are adjacent, since otherwise the matching  $M \cup \{uv\}$  would be greater than  $M$ . Since the graph does not have isolated vertices, we can assign to each vertex  $v \in V \setminus X$  some edge  $e^v$  such that  $v$  is an endpoint of  $e^v$ . By construction,  $F = M \cup \{e^v : v \in V \setminus X\}$  is a set of edges of the graph  $G$  that forms an edge cover of  $G$ . The cardinality of this set is  $|F| = |M| + |X| = |M| + (|V| - 2|M|) = |V| - |M|$ . This shows that the cardinality of a smallest edge cover in the graph  $G$  does not exceed  $|V| - |M|$ .

The above argument can be reversed. Let  $F$  be a smallest edge cover and let  $M$  be a largest matching in the graph  $G$  for which  $M \subseteq F$  holds. Similar to before, let  $X$  be the

set of vertices that are endpoints of some edge in the matching  $M$ . By the definition of matching  $M$ , no two vertices in the set  $X$  are connected by an edge in the set  $F$ . However, since  $F$  is an edge cover, every vertex in the set  $X$  is an endpoint of some edge in the set  $F$ . Since  $F$  is a smallest edge cover, each vertex in the set  $X$  is an endpoint of exactly one edge in the set  $F$ . It also follows from the minimality of the set  $F$  that for every two distinct edges  $e, f \in M$ , no edge in the set  $F$  connects an endpoint of the edge  $e$  with an endpoint of the edge  $f$ . It follows that  $|F| = |M| + (|V| - 2|M|) = |V| - |M|$ . So, the cardinality of the maximum matching in the graph  $G$  is at least  $|V| - |F|$ . If we now denote by  $M'$  an arbitrary maximum matching in the graph  $G$ , we get that  $|F| = |V| - |M| \geq |V| - |M'| = |M'| + (|V| - 2|M'|)$ .

The above considerations show that a smallest edge cover  $F$  can be computed as follows:

- (a) compute a maximum matching  $M$  in the graph  $G$ ,
- (b) set  $F = M$ ,
- (c) for each vertex  $v \in V$  that is not the endpoint of any edge in  $M$ , we add an arbitrary edge to the endpoint of  $v$  to the set  $F$ .

Since the maximum matching problem is polynomially solvable by Edmonds' algorithm, the entire algorithm is polynomial.

#### 4. (Applications of maximum-weight matchings; suggested.)

We mentioned in the lectures that the following problem is polynomial-time solvable:

##### MAXIMUM-WEIGHT MATCHING

**Input:** A graph  $G = (V, E)$ , non-negative weight function on edges,  $w : E \rightarrow \mathbb{Z}_+$ .

**Task:** Find a matching  $M$  in the graph  $G$  of maximum total weight (i.e., a matching maximizing the value of  $w(M) := \sum_{e \in M} w(e)$ ).

Show that the following problem is polynomially solvable by reducing it to the maximum-weight matching problem.

##### COVERING A SET BY A MATCHING

**Input:** A graph  $G = (V, E)$ , a set  $S \subseteq V$ .

**Question:** Is there a matching  $M$  in the graph  $G$  such that every vertex  $v \in S$  is an endpoint of some edge of the matching  $M$ ?

*Hint: Define  $w(e) = |e \cap S|$ . Show that in the graph  $G$  there exists a matching  $M$  such that every vertex  $v \in S$  is an endpoint of some edge in  $M$  if and only if there is a matching in the graph  $G$  with total weight at least  $t$ , for a suitable value of  $t$ .*

*Hint: Define  $w(e) = |e \cap S|$ . Show that in the graph  $G$  there exists a matching  $M$  such that every vertex  $v \in S$  is an endpoint of some edge in  $M$  if and only if there is a matching in the graph  $G$  with total weight at least  $t$ , for a suitable value of  $t$ .*

**Solution:** Let's follow the hint. Let us define  $w(e) = |e \cap S|$ . Let us show that in the graph  $G$  there exists a matching  $M$  such that every vertex  $v \in S$  is an endpoint of some edge in  $M$  if and only if there exists a matching in the graph  $G$  of total weight at least  $|S|$ . It will follow from here that we can solve the COVERING A SET BY A MATCHING problem by computing a maximum-weight matching  $M$  (which we can do in polynomial time) and checking whether  $w(M) \geq |S|$ .

Let us first assume that in the graph  $G$  there is an matching  $M$  such that every vertex  $v \in S$  is an endpoint of some edge in  $M$ . Then

$$\begin{aligned}
w(M) &= \sum_{e \in M} |e \cap S| \\
&= \sum_{e \in M} \sum_{v \in S: v \in e} 1 \\
&= \sum_{v \in S} \sum_{e \in M: v \in e} 1 \\
&= \sum_{v \in S} 1 \\
&= |S|,
\end{aligned}$$

where in the penultimate equality we used the assumption that every vertex  $v \in S$  is an endpoint of some edge in  $M$ . So, in the graph  $G$  there exists an matching of total weight at least  $|S|$ .

Let us now assume that in the graph  $G$  there exists a matching  $M$  of total weight at least  $|S|$ . Then every vertex  $v \in S$  is an endpoint of some edge in  $M$ . Indeed, if there is a vertex  $x \in S$  that is not the endpoint of any edge in  $M$ , then the following holds:

$$\begin{aligned}
w(M) &= \sum_{e \in M} |e \cap S| \\
&= \sum_{e \in M} \sum_{v \in S: v \in e} 1 \\
&= \sum_{v \in S} \sum_{e \in M: v \in e} 1 \\
&\leq \sum_{v \in S \setminus \{x\}} \sum_{e \in M: v \in e} 1 \\
&\leq \sum_{v \in S \setminus \{x\}} 1 \\
&= |S| - 1,
\end{aligned}$$

a contradiction.

5. (*Matchings in claw-free graphs; part (a) is suggested, part (b) more challenging.*)

A matching in a graph  $G$  is *perfect* if every vertex of the graph  $G$  is an endpoint of some edge in  $M$ . A *claw* is the complete bipartite graph  $K_{1,3}$ .<sup>1</sup> A graph  $G$  is *claw-free* if it has no induced subgraph isomorphic to a claw. In other words, for each vertex  $v \in V(G)$ , the neighborhood of  $v$  does not contain three pairwise non-adjacent vertices.

Let  $G$  be a claw-free connected graph with an even number of vertices.

- (a) Let  $s \in V(G)$  and let  $T$  be an arbitrary BFS tree in the graph  $G$ , rooted at vertex  $s$ . Let  $x$  and  $y$  be two vertices of the graph  $G$  that have a common parent  $z \neq s$  in the tree  $T$ . Show that the vertices  $x$  and  $y$  are adjacent.

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<sup>1</sup>That is, it has vertex set  $V = A \cup B$ , where  $A \cap B = \emptyset$ ,  $|A| = 1$ ,  $|B| = 3$ , the unique vertex in the set  $A$  is adjacent to all vertices in the set  $B$ , and there are no other edges.

**Solution:** Let us prove the statement by contradiction. Assume that the vertices  $x$  and  $y$  are not adjacent. Since  $z \neq s$ , the vertex  $z$  has a parent  $u$  in the tree  $T$ . Since  $T$  is a BFS tree, vertex  $u$  is not adjacent to either vertex  $x$  or vertex  $y$ . But this means that the vertices  $\{x, y, z, u\}$  induce a claw in the graph  $G$ , which is a contradiction.

- (b) Using part (a), show that a perfect matching exists in the graph  $G$ .

(Note: without using part (a) one can show a stronger result, that the last edge of a longest path in the graph  $G$  belongs to some perfect matching.)

**Solution:** We use the property from part (a) and induction by  $n = |V(G)|$  to construct a perfect matching in the graph  $G$  from the leaves of the tree  $T$  towards the root. If  $n = 2$ , the graph is a complete graph with two vertices and the only edge in it forms a perfect matching. For  $n > 2$ , consider a BFS tree  $T$  with root  $s$  and the set of vertices  $U \subseteq V(G)$  at the maximum distance from the vertex  $s$ .

Let us first assume that there are two adjacent vertices  $x, y$  in the set  $U$ . Since the graph  $G - x - y$  is connected and claw-free, it has, by the induction assumption, a perfect matching  $M$ . Then  $M \cup \{xy\}$  is a perfect matching in the graph  $G$ .

So we can assume that no two vertices in the set  $U$  are adjacent. Let  $x \in U$  and let  $z$  be the parent of  $x$  in tree  $T$ . If  $z = s$ , then  $U = V(G) \setminus \{s\}$  and consequently  $|N(s)| = n - 1 \geq 3$ , so  $G$  contains an induced claw, a contradiction. So  $z \neq s$  and from part (a) we derive that there is no vertex  $y \in U \setminus \{x\}$  such that the vertices  $x$  and  $y$  in the tree  $T$  have a common parent. This means that the graph  $G - x - z$  is connected (since it has a spanning tree  $T - x - z$ ). Since it is claw-free, it has, by induction assumption, some perfect matching  $M$ . Then  $M \cup \{xz\}$  is a perfect matching in the graph  $G$ .